
Turn in 4

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MA 542 Section A1

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Note: there is a lot of explicit matrix computation involved in this assignment. I have included all computations at the end just incase you want to see them, but I wasn't sure and didn't want to make it too messy so like yeah.

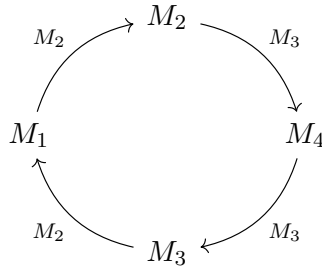
For simplicity, denote

$$M_1 := \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad M_2 := \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad M_3 := \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad M_4 := \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Also let 1_R denote the identity matrix. Explicit computation yields that

$$M_1 M_2 = M_2, \quad M_3 M_2 = M_4, \quad M_4 M_3 = M_3, \quad M_2 M_3 = M_1.$$

This can be visualized with the diagram:



In some cases the operation is left-multiplication, and in others the operation is right-multiplication. The important thing to note is that if there exists some $i \in \{1, \dots, 4\}$ with $M_i \in I$, then it follows that $M_i \in I$ for all i , because I is an ideal and thus has the absorption property, and all M_i can be represented as a product in R on a single M_i by the above diagram. I will now show the ideal contains at least one of these.

Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in I$$

be nonzero. Note that $A = aM_1 + bM_2 + cM_3 + dM_4$. Since A is nonzero, at least one element in $\{a, b, c, d\}$ must be nonzero. If precisely one element is nonzero, then we have some coefficient $k \in \mathbb{R}$ such that

$$A = kM_i$$

for some i . Since I is an ideal, $A \in I$, and $\frac{1}{k}1_R \in R$, we have $\frac{1}{k}1_R A = \frac{k}{k}1_R M_i = M_i$ is an element of the ideal.

Now suppose that more than one coefficient is nonzero. In particular, we must have at least two coefficients are nonzero. Consider the following three cases:

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- **Case 1:** $a, b \neq 0$, and $c, d = 0$. That is,

$$A = aM_1 + bM_2.$$

Explicit computation shows that $M_1M_4 = 0$ and $M_2M_4 = M_2$. It follows that

$$AM_4 = (aM_1 + bM_2)M_4 = aM_1M_4 + bM_2M_4 = bM_2.$$

Here, I use some facts about vector spaces, such as $a0_R = 0_R$ for all scalars a . Since I is an ideal and $a \in I$, we have $AM_4 = bM_2 \in I$. By the same logic, we also have

$$\frac{1}{b}1_R(bM_2) = 1_RM_2 = M_2 \in I.$$

- **Case 2:** $c, d \neq 0$, and $a, b = 0$. That is,

$$A = cM_3 + dM_4.$$

Explicit computation yields that $M_3M_1 = M_3$ and $M_4M_1 = 0$. Therefore, we have

$$AM_1 = (cM_3 + dM_4)M_1 = cM_3M_1 + dM_4M_1 = cM_3.$$

Since $A \in I$ and I is an ideal, $AM_1 = cM_3 \in I$. Similarly,

$$\frac{1}{c}1_R(cM_3) = M_3 \in I.$$

- **Case 3:** Any other possible scenario. We must have either a , b , or both are nonzero. By explicit computation, we have

$$M_1M_1 = M_1, \quad M_1M_2 = M_2, \quad M_1M_3, M_1M_4 = 0.$$

Since left multiplication fixes M_1, M_2 and annihilates M_3, M_4 , we can take $c, d = 0$. If only one of the coefficients a, b are nonzero, then we have the corresponding $M_i \in I$ by previous logic. If both are nonzero, apply the same logic as in case 1.

These are the only cases, since $A \neq 0$. Therefore, $M_1, M_2, M_3, M_4 \in I$.

Since I is an ideal, it must be a subring, and thus closed under addition. We thus have

$$M_1 + M_4 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = 1_R \in I.$$

Let $M \in \mathcal{M}_2(\mathbb{R})$ be any matrix. Then since $1_R \in I$, and I is an ideal, we have that $1_RM = M \in I$. Therefore, $\mathcal{M}_2(\mathbb{R}) \subseteq I$. Since I is an ideal, we have $I \subseteq \mathcal{M}_2(\mathbb{R})$, and thus $I = \mathcal{M}_2(\mathbb{R})$.

$$i\hbar \frac{d}{dt} |\Psi(t)\rangle = \hat{\mathbf{H}} |\Psi(t)\rangle.$$